

Reading Notes

Reduced Density Fluctuations via Antialigning in Active Matter

Reduced Density Fluctuations via Antialigning in Active Matter

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Notes compiled

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Abstract

This document provides detailed reading notes on the theoretical analysis section of the paper by Boltz and Ihle, published in Physical Review E in 2025¹. The paper investigates how antialigning interactions reduce long-range density fluctuations in active matter systems, and may even lead to the formation of hyperuniform states. These notes focus on filling in the theoretical derivation steps that were omitted or simplified in the original article, including: construction of the one-dimensional lattice model and its master equation, detailed derivation of the Poisson representation, construction of the Fokker-Planck equation, derivation of the Langevin equations, and analytical calculation of the structure factor $S(k)$. Finally, numerical simulation results for a two-dimensional Vicsek-like model are briefly discussed.

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1 Introduction

1.1 Research Background

Active matter refers to particulate systems driven out of equilibrium at the particle scale¹. Such systems exhibit many collective phenomena that are inaccessible to equilibrium physics, the most famous example being long-range orientational order in two dimensions³.

In general, activity (particularly the self-propulsion of particles) is a means of establishing long-range correlations in dilute systems, even when interactions are short-ranged. This property holds even in the absence of global spatial order, orientational order, or any other type of order. Taking this to the extreme: even when the nonequilibrium steady state is characterized by a uniform **single-particle distribution function**, correlations between particles remain crucial for understanding the behavior of the system.

Key Physical Picture

These correlations quantify the statistical deviation of positional configurations from realizations of a simple **homogeneous Poisson point process**. Even in the absence of any macroscopic order, long-range correlations in active systems can still lead to significant corrections to statistical fluctuations.

1.2 Hyperuniformity

Hyperuniform configurations⁴ are a special class of disordered point patterns in which the particle number fluctuations of large subsystems exhibit **anomalous scaling behavior**:

$$\Delta N \sim r^\beta \quad (r \rightarrow \infty), \quad d-1 < \beta < d \quad (1)$$

where d is the spatial dimension. This stands in sharp contrast to the bulk law of ordinary Poisson processes, $\Delta N \sim \sqrt{N} \sim r^{d/2}$.

Complementary definition: Hyperuniformity can also be defined via the static structure factor:

$$S(\mathbf{k}) = \langle \delta \tilde{\rho}(\mathbf{k}) \delta \tilde{\rho}(-\mathbf{k}) \rangle \quad (2)$$

When $S(\mathbf{k} \rightarrow \mathbf{0}) \rightarrow 0$, the system is in a hyperuniform state.

1.3 Antialigning Interactions

Antialigning orientational interactions are less studied than aligning interactions, but exist in certain experiments¹. For dilute systems, density fluctuations cause particles to interact and typically produce opposite orientations. Therefore, **center-of-mass motion** is suppressed (though not completely forbidden by conservation laws), and successive interactions of the same particle are also suppressed.

Core Intuition

The core intuition of this paper is: in systems of self-driven particles with short-range antialigning interactions, density fluctuations are “dissolved” in a nonlocal manner. Similar to Manna processes or adsorbed-state systems with center-of-mass conservation, the correlation mechanism in antialigning systems can effectively suppress large-scale density fluctuations.

1.4 Main Contributions of the Paper

The main contributions of this paper include:

1. **One-dimensional analytical model:** A one-dimensional active lattice gas model is constructed and analyzed rigorously using the **Poisson representation**
2. **Fluctuating hydrodynamics:** Fluctuating hydrodynamic equations for the Poisson fields are derived
3. **Discovery of imaginary noise:** **Imaginary noise** describing non-Poissonian statistics is discovered, marking the non-Poissonian character of particle number fluctuations
4. **Analytical calculation of the structure factor:** The nontrivial structure factor $S(k)$ is calculated analytically
5. **Two-dimensional numerical verification:** Theoretical predictions are verified through simulations of an off-lattice Vicsek-like model

These notes focus on filling in the theoretical derivation details omitted in the original article, particularly:

- Construction of the master equation and complete derivation of the Poisson representation
- Detailed construction of the Fokker-Planck equation
- Derivation process of the Langevin equations
- Analytical calculation of the structure factor

2 One-Dimensional Pedagogical Model

2.1 Model Definition

We consider a one-dimensional lattice populated with two species of particles:

- (+) particles: move rightward to $i + 1$ at a constant rate λ
- (−) particles: move leftward to $i - 1$ at a constant rate λ

Let $n_i^\pm$ denote the number of species $\pm$ particles at lattice site i . The dynamics of the system consists of two processes:

Model Dynamics

1. **Streaming:** Particles move from i to $i \pm 1$ at rate λ
2. **Conversion:** Particles convert from $+$ to $-$ (or vice versa) at rate $r_\pm(n_i^+, n_i^-)$

The choice of conversion rates reflects the physics of antialigning interactions. In this paper, the simplest nonlinear form is chosen (ignoring self-interactions):

$$r_\pm(n^+, n^-) = r_0 + (n^\pm - 1)r_1 \quad (3)$$

where r_0 is the “spontaneous conversion rate” and r_1 is the antialigning interaction strength associated with the local magnetization.

Remark (Note on the form of conversion rates). Here we require r_1 to be much larger than r_0 , because we are focusing on conversions induced by local magnetization. Keeping only terms up to $O(n^+, n^-)$ in the conversion rate is crucial—higher-order terms would lead to derivatives higher than second order in the Fokker-Planck equation, which cannot be solved directly via the corresponding Langevin equation.

2.2 Master Equation

Let $P(\{n_i^+, n_i^-\})$ be the probability distribution of finding a specific set of occupation numbers. The master equation reads:

$$\partial_t P = -S_{\text{out}}^\lambda + S_{\text{in}}^\lambda - C_{\text{out}}^r + C_{\text{in}}^r \quad (4)$$

The individual terms represent:

- S_{out}^λ : Probability outflow due to streaming
- S_{in}^λ : Probability inflow due to streaming
- C_{out}^r : Probability outflow due to conversion
- C_{in}^r : Probability inflow due to conversion

2.2.1 Explicit Expressions for Probability Currents

Streaming terms (outflow):

$$S_{\text{out}}^\lambda = \sum_i [\lambda n_i^+ + \lambda n_i^-] P \quad (5)$$

$$C_{\text{out}}^r = \sum_i [r_+ n_i^+ + r_- n_i^-] P \quad (6)$$

Streaming terms (inflow, contributions from left and right neighbors):

$$S_{\text{in}}^\lambda = \sum_i \left[\lambda (n_{i-1}^+ + 1) P(\dots, n_{i-1}^+ + 1, n_i^+ - 1, \dots) \right. \\ \left. + \lambda (n_{i+1}^- + 1) P(\dots, n_i^- - 1, n_{i+1}^- + 1, \dots) \right] \quad (7)$$

Conversion terms (inflow):

$$C_{\text{in}}^r = \sum_i \left[r_+ (n_i^+ + 1, n_i^- - 1) (n_i^+ + 1) P(\dots, n_i^+ + 1, n_i^- - 1, \dots) \right. \\ \left. + r_- (n_i^+ - 1, n_i^- + 1) (n_i^- + 1) P(\dots, n_i^+ - 1, n_i^- + 1, \dots) \right] \quad (8)$$

2.3 Poisson Representation

Directly solving the master equation is infeasible because for interacting systems (i.e., nontrivial conversion rates), the right-hand side is nonlinear. Therefore, we adopt the **Poisson representation**², a method that transforms a discrete stochastic process into a continuous one.

2.3.1 Core Idea

Core Idea of the Poisson Representation

Replace the discrete occupation numbers $n_i^\pm$ (a stochastic process on positive integers) with contin-

uous Poisson rates α_i, β_i (corresponding to $+$ and $-$ particles, respectively). These rates serve as parameters of Poisson processes that then generate the occupation numbers $n_i^\pm$.

Statistical moments are obtained by averaging over realizations of the Poisson process for fixed fields, as well as over the distribution of these fields.

The relationship between the probability distribution $P(\{n_i^+, n_i^-\})$ and the quasiprobability distribution $f(\{\alpha_i, \beta_i\})$ is:

$$P = \int \prod_i d\alpha_i d\beta_i e^{-\alpha_i} \frac{\alpha_i^{n_i^+}}{n_i^+!} \cdot e^{-\beta_i} \frac{\beta_i^{n_i^-}}{n_i^-!} f \quad (9)$$

2.3.2 Mismatch Between General Discrete Processes and Poisson Processes

Due to a “mismatch” between the generality of discrete processes and local Poisson processes, a higher level of abstraction is required:

1. The local rates α_i, β_i are **complex numbers**
2. The distribution f is actually a **quasidistribution**, as it can also be locally negative

These extensions are not unphysical, because the Poisson fields are auxiliary mathematical objects that are not directly observable. **Observable** are the moments of the physical process, which are positive and real.

Remark (Intuitive understanding of imaginary numbers). The contribution of imaginary numbers to the Poisson field can be viewed as a manifestation of deviations from Poisson statistics, which explicitly couples the mean and variance of the random variables.

Compare a Poisson variable n with rate λ and a variable m with rate $\lambda + i\xi$, where ξ is a zero-mean random variable with $\langle \xi^2 \rangle = c$. For a Poisson process, we have:

$$\langle n \rangle = \langle n^2 \rangle_c = \lambda$$

But for m :

$$\langle m \rangle = \lambda, \quad \langle m^2 \rangle_c = \lambda - c^2$$

Imaginary noise leads to **narrowing** of the microscopic distribution, while real noise leads to **widening**.

2.3.3 Replacement Rules: From Master Equation to Fokker-Planck Equation

By substituting the Poisson representation into the master equation and using integration by parts, we obtain the following replacement rules. Below, arguments in f and P that are consistent with the aforementioned default parameters are omitted.

Replacement rules for linear terms (streaming and spontaneous conversion)

Outflow term (rightward motion):

$$n_i^+ P \rightarrow (\alpha_i f - \partial_{\alpha_i} \alpha_i f) \quad (10)$$

Outflow term (leftward motion):

$$n_i^- P \rightarrow (\beta_i f - \partial_{\beta_i} \beta_i f) \quad (11)$$

Inflow term (from the left):

$$(n_{i-1}^+ + 1)P(\dots, n_{i-1}^+ + 1, n_i^+ - 1, \dots) \rightarrow (\alpha_{i-1} - \partial_{\alpha_i} \alpha_{i-1})f \quad (12)$$

Inflow term (from the right):

$$(n_{i+1}^- + 1)P(\dots, n_i^- - 1, n_{i+1}^- + 1, \dots) \rightarrow (\beta_{i+1} - \partial_{\beta_i} \beta_{i+1})f \quad (13)$$

Replacement rules for nonlinear terms (conversion process)

Conversion from (+) to (-):

$$n_i^+(n_i^+ - 1)P \rightarrow (1 - \partial_{\alpha_i})^2 \alpha_i^2 f \quad (14)$$

Conversion from (-) to (+):

$$n_i^-(n_i^- - 1)P \rightarrow (1 - \partial_{\beta_i})^2 \beta_i^2 f \quad (15)$$

Conversion of (+) particles induced by (-) particles:

$$(n_i^- + 1)P(\dots, n_i^+ - 1, n_i^- + 1, \dots) \rightarrow (-1 + \partial_{\alpha_i})(-1 + \partial_{\beta_i}) \alpha_i^2 f \quad (16)$$

Conversion of (-) particles induced by (+) particles:

$$(n_i^+ + 1)P(\dots, n_i^+ + 1, n_i^- - 1, \dots) \rightarrow (-1 + \partial_{\alpha_i})(-1 + \partial_{\beta_i}) \beta_i^2 f \quad (17)$$

2.3.4 Detailed Derivation: Obtaining the Replacement Rules

Taking Eq. (10) as an example, we show the detailed derivation:

$$\begin{aligned} n_i^+ P &= \int \prod_j d\alpha_j d\beta_j n_i^+ \frac{e^{-\alpha_i} \alpha_i^{n_i^+}}{n_i^+!} \cdot \frac{e^{-\beta_i} \beta_i^{n_i^-}}{n_i^-!} f \\ &= \int \prod_j d\alpha_j d\beta_j \frac{e^{-\alpha_i} \alpha_i^{n_i^+}}{(n_i^+ - 1)!} \cdot \frac{e^{-\beta_i} \beta_i^{n_i^-}}{n_i^-!} f \\ &= \int \prod_j d\alpha_j d\beta_j \frac{e^{-\alpha_i} \alpha_i^{n_i^+}}{n_i^+!} \cdot \frac{e^{-\beta_i} \beta_i^{n_i^-}}{n_i^-!} \alpha_i f \\ &\quad - \int \prod_j d\alpha_j d\beta_j (-\partial_{\alpha_i}) \left[e^{-\alpha_i} \frac{\alpha_i^{n_i^+}}{n_i^+!} \right] \cdot \frac{e^{-\beta_i} \beta_i^{n_i^-}}{n_i^-!} \alpha_i f \\ &= \text{form of } P \text{ after } n_i^+ \text{ is replaced} \end{aligned}$$

After integration by parts (assuming boundary terms vanish), the replacement rule is obtained.

For nonlinear terms, e.g., Eq. (14):

$$\begin{aligned}
n_i^+(n_i^+ - 1)P &= \int d\alpha_i n_i^+(n_i^+ - 1) \frac{e^{-\alpha_i} \alpha_i^{n_i^+}}{n_i^+!} f \\
&= \int d\alpha_i \frac{e^{-\alpha_i} \alpha_i^{n_i^+}}{(n_i^+ - 2)!} f \\
&= \int d\alpha_i (1 - \partial_{\alpha_i})^2 \left[\frac{e^{-\alpha_i} \alpha_i^{n_i^+}}{n_i^+!} \right] \alpha_i^2 f \\
&\rightarrow (1 - \partial_{\alpha_i})^2 \alpha_i^2 f
\end{aligned}$$

2.4 Fokker-Planck Equation

Collecting the above replacement rules, we obtain the Fokker-Planck equation for the quasiprobability density $f(\{\alpha_i, \beta_i\})$:

$$\partial_t f = F_\lambda + F_r + J \quad (18)$$

where the individual terms are:

2.4.1 Streaming Contribution F_λ

$$\begin{aligned}
F_\lambda &= \lambda \sum_i \left[(\alpha_{i-1} - \alpha_i) + \partial_{\alpha_i} (\alpha_i - \alpha_{i-1}) \right] f \\
&\quad - \lambda \sum_i \left[(\beta_i - \beta_{i+1}) + \partial_{\beta_i} (\beta_{i+1} - \beta_i) \right] f
\end{aligned} \quad (19)$$

2.4.2 Conversion Contribution F_r

$$\begin{aligned}
F_r &= r_0 \sum_i \left[\partial_{\alpha_i} (\alpha_i - \beta_i) - \partial_{\beta_i} (\alpha_i - \beta_i) \right] f \\
&\quad + r_1 \sum_i \left[\partial_{\alpha_i} - \partial_{\beta_i} \right] (\alpha_i + \beta_i) (\alpha_i - \beta_i) f
\end{aligned} \quad (20)$$

2.4.3 Noise Term J

$$J = -r_1 \sum_i \left[\partial_{\alpha_i}^2 \alpha_i^2 + \partial_{\beta_i}^2 \beta_i^2 \right] f \quad (21)$$

The last line contains second-derivative terms, which manifest as **noise** in the corresponding Langevin equation. Note that the coefficient of this noise term is **negative**, which ultimately leads to the emergence of imaginary noise.

2.5 Deterministic Equations of Motion

From the Fokker-Planck equation, we can read off the deterministic equations (ignoring the noise term). Changing the notation from λ to v (for consistency with tradition), we obtain:

$$\partial_t \alpha|_{\text{det}} = -v \nabla \alpha - r_0 (\alpha - \beta) - r_1 (\alpha + \beta) (\alpha - \beta) \quad (22)$$

$$\partial_t \beta|_{\text{det}} = v \nabla \beta + r_0 (\alpha - \beta) + r_1 (\alpha + \beta) (\alpha - \beta) \quad (23)$$

Note that here ∇ is shorthand for nearest-neighbor differences, and the **continuous limit has not yet been taken**.

2.6 Noise Analysis

To determine the noise term, we write the Fokker-Planck equation in standard form:

$$\partial_t f = - \sum_i \partial_{x_i} (\mu_i f) + \sum_{i,j} \partial_{x_i} \partial_{x_j} [D_{ij} f] \quad (24)$$

The diffusion matrix D_{ij} is:

$$D_{ij} = -r_1 \begin{pmatrix} \alpha_i^2 & 0 \\ 0 & \beta_i^2 \end{pmatrix} \quad (25)$$

Introducing the variable transformation:

$$\tilde{\rho}_i = \alpha_i + \beta_i \quad (\text{total density}), \quad \tilde{m}_i = \alpha_i - \beta_i \quad (\text{magnetization}) \quad (26)$$

The diffusion matrix becomes:

$$D_{ij} = -r_1 \begin{pmatrix} \frac{\tilde{\rho}^2 + \tilde{m}^2}{4} & \frac{-2\tilde{\rho}\tilde{m}}{4} \\ \frac{-2\tilde{\rho}\tilde{m}}{4} & \frac{\tilde{\rho}^2 + \tilde{m}^2}{4} \end{pmatrix} \quad (27)$$

2.6.1 Matrix Square Root and Imaginary Noise

Translating the Fokker-Planck equation into Langevin equations requires computing the matrix square root $B = \sqrt{2D}$. In the physically relevant limit $\tilde{\rho} \approx \rho_0$, $\tilde{m} \approx 0$, to zeroth order in $\tilde{m}$:

$$B = i\rho_0 \sqrt{\frac{r_1}{2}} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (28)$$

Here the factor $i = \sqrt{-1}$ appears! This is the hallmark result of the Poisson representation.

Physical Meaning of Imaginary Noise

The two eigenvalues are:

$$\lambda_\rho = 0, \quad \lambda_m = i\sqrt{2r_1\rho_0}$$

$\lambda_\rho = 0$ means that **the density field has no noise**, while the magnetization field has imaginary noise.

This does not imply that any observable is imaginary. The appearance of imaginary noise signals a deviation from Poisson statistics—in the Poisson representation, **only complex fields can describe non-Poissonian statistics**, because the mean and variance of a Poisson distribution must be equal, and to break this equality, the imaginary part of the field must be nonzero.

2.7 Langevin Equations

Summarizing the above analysis, the coupled Langevin equations for the density and magnetization fields are:

Langevin Equations

$$\partial_t \tilde{\rho} = -v \nabla \tilde{m} \quad (29)$$

$$\partial_t \tilde{m} = -v \nabla \tilde{\rho} - r_0 \tilde{m} - r_1 \tilde{\rho} \tilde{m} + i \sqrt{2r_1 \tilde{\rho}} \xi \quad (30)$$

where ξ is Gaussian uncorrelated noise satisfying:

$$\langle \xi \rangle = 0, \quad \langle \xi(t) \xi(t') \rangle = \delta(t - t') \quad (31)$$

Remark (Term-by-term physical interpretation). **Density equation (29):**

- $\partial_t \tilde{\rho}$: Local time rate of change of the total density field.
- $-v \nabla \tilde{m}$: **Density current driven by magnetization gradient**. Since (+) particles move rightward and (−) particles move leftward, if there are more rightward-moving particles to the right than to the left at a given location (i.e., $\nabla \tilde{m} > 0$), the density at that location decreases, and vice versa. The negative sign reflects the net transport of particles from regions of high magnetization to regions of low magnetization.

Magnetization equation (30):

- $\partial_t \tilde{m}$: Time rate of change of the magnetization (i.e., the difference between the number of rightward and leftward particles).
- $-v \nabla \tilde{\rho}$: **Magnetization current driven by density gradient**. If the density increases to the right ($\nabla \tilde{\rho} > 0$), more rightward-moving particles flow in from the left, causing the local magnetization to decrease.
- $-r_0 \tilde{m}$: **Magnetization relaxation due to spontaneous conversion**. r_0 is the spontaneous flipping rate, which causes (+) and (−) particles to convert into each other, driving the system toward an unpolarized state ($\tilde{m} = 0$); this is an exponential decay term.
- $-r_1 \tilde{\rho} \tilde{m}$: **Nonlinear relaxation due to antialigning interaction**. This is the most crucial term in this paper: the stronger the local magnetization, the faster particles flip to the opposite direction due to the “antialigning” mechanism; moreover, this effect is proportional to the local total density (the higher the density, the more frequent the collisions/interactions). This term is the key to suppressing magnetization and, in turn, suppressing density fluctuations.
- $i \sqrt{2r_1 \tilde{\rho}} \xi$: **Imaginary noise term**. This is the hallmark result of the Poisson representation. The discreteness and nonlinearity of the conversion process lead to intrinsic stochastic fluctuations in the magnetization dynamics; the negative definiteness of the diffusion matrix D makes the noise coefficient purely imaginary, meaning that the system exhibits **sub-Poissonian** statistics (fluctuations narrower than the Poisson distribution). Note that there is no noise at all in the density equation, which is an important signature of the strong suppression of fluctuations in this model.

Remark (Connection to existing literature). The deterministic version of this set of equations (setting $r_0 \equiv 0$) has the same structure as the closed hydrodynamics of antialigning particles¹ in one dimension. General Toner-Tu-type approaches⁶ also encompass these dynamics, but do not make specific predictions for r_0 and r_1 .

2.8 Analytical Calculation of the Structure Factor

2.8.1 Linearization and Simplification

Consider small perturbations around the uniform unpolarized state:

$$\tilde{\rho} = \rho_0 + \delta\tilde{\rho}, \quad \tilde{m} \text{ treated as a small quantity} \quad (32)$$

Approximating the multiplicative noise as additive noise (ignoring the $\delta\tilde{\rho}\xi$ term), we obtain a closed second-order equation:

$$\partial_t^2 \delta\tilde{\rho} = v^2 \nabla^2 \delta\tilde{\rho} - (r_0 + r_1 \rho_0) \partial_t \delta\tilde{\rho} - i\sqrt{2r_1 \rho_0} v \nabla \xi \quad (33)$$

2.8.2 Overdamped Approximation

Write the **deterministic** part of the second-order equation (33) (ignoring noise for now) in Fourier space, setting $\nabla^2 \rightarrow -k^2$:

$$\partial_t^2 \delta\tilde{\rho}_k + \gamma \partial_t \delta\tilde{\rho}_k + \omega_0^2 \delta\tilde{\rho}_k = 0 \quad (34)$$

Notation: Meaning of $\delta\tilde{\rho}_k$

Recall Eq. (32), where we wrote the total density as the sum of a uniform background and a small perturbation: $\tilde{\rho} = \rho_0 + \delta\tilde{\rho}$. The subscript k denotes the **wavenumber component** after Fourier transforming the spatial coordinate:

$$\delta\tilde{\rho}_k(t) = \int dx e^{-ikx} \delta\tilde{\rho}(x, t) \quad (\text{continuous}) \quad \text{or} \quad \delta\tilde{\rho}_k(t) = \sum_j e^{-ikj} \delta\tilde{\rho}_j(t) \quad (\text{discrete lattice})$$

Thus $\delta\tilde{\rho}_k$ is the **Fourier mode amplitude** of the density fluctuation at wavenumber k . When taking time derivatives, the spatial derivative ∇ is replaced by ik (or $\nabla^2 \rightarrow -k^2$), thereby reducing the partial differential equation to an ordinary differential equation for each k .

This is the standard form of a **damped harmonic oscillator**, where:

- $\gamma = r_0 + r_1 \rho_0$ (damping coefficient, from spontaneous conversion and antialigning interaction)
- $\omega_0^2 = v^2 k^2$ (square of the natural frequency, from the directed motion of particles)

The characteristic equation of the damped harmonic oscillator is $\lambda^2 + \gamma\lambda + \omega_0^2 = 0$, yielding two eigenvalues:

$$\lambda_{\pm} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4\omega_0^2}}{2} \quad (35)$$

Overdamped condition: When $\gamma^2 \gg 4\omega_0^2$, the quantity under the square root is positive, and both eigenvalues are **real and negative**; the system does not oscillate but decays directly. In this model:

$$\gamma^2 = (r_0 + r_1 \rho_0)^2, \quad 4\omega_0^2 = 4v^2 k^2$$

Since γ is a constant while $\omega_0 \propto k$, in the **large-scale limit** $k \rightarrow 0$, the condition is automatically satisfied! This is the fundamental reason why the system is overdamped at large scales: the rapid conversion between particles (γ) smooths out the oscillatory tendency induced by streaming.

Remark (Why expand the square root? On what approximation is this based?). Without expansion, the eigenvalues (35) are wrapped inside a square root, and we **cannot directly read off** the scaling behavior of the two modes. The core of the overdamped limit is $\gamma^2 \gg 4\omega_0^2$, i.e.,

$$\epsilon \equiv \frac{4\omega_0^2}{\gamma^2} = \frac{4v^2k^2}{(r_0 + r_1\rho_0)^2} \ll 1$$

This is a **small parameter**. Rewrite the square root as

$$\sqrt{\gamma^2 - 4\omega_0^2} = \gamma\sqrt{1 - \epsilon}$$

and Taylor expand $\sqrt{1 - \epsilon}$ (to first order):

$$\sqrt{1 - \epsilon} = 1 - \frac{\epsilon}{2} + O(\epsilon^2) \approx 1 - \frac{2\omega_0^2}{\gamma^2}$$

In this way the square root is **linearized**, and upon substitution the k dependence of the two eigenvalues becomes transparent: one is proportional to $-\gamma$ (fast mode), and the other is proportional to $-k^2$ (slow mode). If the square root were kept, subsequent scaling analysis of the structure factor and diffusion coefficient would be obstructed.

Expanding the square root in the $k \rightarrow 0$ limit:

$$\sqrt{\gamma^2 - 4\omega_0^2} \approx \gamma \left(1 - \frac{2\omega_0^2}{\gamma^2} \right) = \gamma - \frac{2\omega_0^2}{\gamma}$$

Substituting into Eq. (35), we obtain two modes:

$$\lambda_+ = \frac{-\gamma + (\gamma - 2\omega_0^2/\gamma)}{2} = -\frac{\omega_0^2}{\gamma} = -\frac{v^2k^2}{r_0 + r_1\rho_0} \quad (36)$$

$$\lambda_- = \frac{-\gamma - (\gamma - 2\omega_0^2/\gamma)}{2} = -\gamma + \frac{\omega_0^2}{\gamma} \approx -(r_0 + r_1\rho_0) \quad (37)$$

- λ_- is the **fast mode** ($\sim -\gamma$), decaying to zero in extremely short times via $e^{\lambda_- t}$.
- λ_+ is the **slow mode** ($\sim -k^2$), determining the long-time behavior of the system.

Therefore, the long-time behavior is dominated by the slow mode, i.e.:

$$\boxed{\mu = \lambda_+ = -\frac{v^2k^2}{r_0 + r_1\rho_0}} \quad (38)$$

This gives the diffusive scaling $\sim e^{-Dk^2t}$, where the diffusion coefficient is:

$$\boxed{D = \frac{v^2}{r_0 + r_1\rho_0}} \quad (39)$$

Remark (Another intuitive understanding of overdamping). One can also see the essence of overdamping directly from the second-order equation (33): the coefficient of the damping term $\gamma\partial_t\delta\tilde{\rho}$ is $\gamma = r_0 + r_1\rho_0$, which at $k \rightarrow 0$ is much larger than the effective coefficient of the inertial term $\partial_t^2\delta\tilde{\rho}$ (which can be taken as 1). Therefore, we can **adiabatically neglect the inertial term** $\partial_t^2\delta\tilde{\rho} \approx 0$, immediately obtaining:

$$0 \approx -v^2k^2\delta\tilde{\rho}_k - (r_0 + r_1\rho_0)\partial_t\delta\tilde{\rho}_k \implies \partial_t\delta\tilde{\rho}_k = -\frac{v^2k^2}{r_0 + r_1\rho_0}\delta\tilde{\rho}_k$$

This is completely consistent with Eq. (38). Physically, this means that the magnetization (fast variable) is **slaved** by the density (slow variable), and the system relaxes by diffusion without oscillation.

2.8.3 First-Order Langevin Equation

Since the deterministic dynamics is overdamped at large scales, we can neglect the inertial term and reduce everything to a first-order Langevin equation (or noisy diffusion equation):

$$\partial_t \delta \tilde{\rho}_k = -Dk^2 \delta \tilde{\rho}_k + ic\eta_k \quad (40)$$

where:

$$c = v \sqrt{\frac{2r_1 \rho_0}{r_0 + r_1 \rho_0}} \quad (41)$$

2.8.4 Definition of Correlated Noise

To correctly understand the effective noise statistics in Fourier space, note that $\nabla \xi$ is actually $\xi_{i+1} - \xi_i$. For this purpose, we introduce the zero-drift correlated noise $\eta_i = \eta(i)$:

$$\langle \eta_i(t) \eta_j(t') \rangle = \delta(t - t') \times \begin{cases} 2, & i = j \\ -1, & i = j \pm 1 \\ 0, & \text{otherwise} \end{cases} \quad (42)$$

In Fourier space:

$$\langle \eta_k(t) \eta_{k'}(t') \rangle = \delta(k + k') \delta(t - t') \cdot 2[1 - \cos(k)] \quad (43)$$

2.8.5 Final Expression for the Structure Factor

Using the above correlated noise to solve Eq. (40), we obtain the structure factor in the long-time limit. Note that the structure factor can be calculated from fluctuations of the correlated Poisson fields⁵:

$$N_S(k) = \langle \rho_k \rho_{-k} \rangle = \rho_0 + \langle \delta \tilde{\rho}_k \delta \tilde{\rho}_{-k} \rangle \quad (44)$$

Remark (Why can $N_S(k)$ be written as $\rho_0 + \langle \delta \tilde{\rho}_k \delta \tilde{\rho}_{-k} \rangle$?). This is a standard identity of the Poisson representation. The physical occupation numbers $n_i = n_i^+ + n_i^-$ are Poisson variables with rates α_i, β_i . For Poisson processes we have:

$$\langle n_i^\pm n_j^\pm \rangle = \langle \alpha_i \alpha_j \rangle + \delta_{ij} \langle \alpha_i \rangle, \quad \langle n_i^+ n_j^- \rangle = \langle \alpha_i \beta_j \rangle$$

Therefore, the correlation of the total particle number is:

$$\begin{aligned} \langle n_i n_j \rangle &= \langle (n_i^+ + n_i^-)(n_j^+ + n_j^-) \rangle \\ &= \langle (\alpha_i + \beta_i)(\alpha_j + \beta_j) \rangle + \delta_{ij} \langle \alpha_i + \beta_i \rangle \\ &= \langle \tilde{\rho}_i \tilde{\rho}_j \rangle + \delta_{ij} \rho_0 \end{aligned}$$

The second term $\delta_{ij} \rho_0$ is precisely the **self-correlation** contribution from Poisson statistics (fluctuations in the particle number at the same lattice site follow a Poisson distribution, with variance equal to the mean). Transforming to Fourier space and considering $k \neq 0$ modes (for which $\tilde{\rho}_k = \delta \tilde{\rho}_k$, because the uniform background ρ_0 contributes only at $k = 0$), we obtain Eq. (44).

Detailed calculation of the structure factor

The solution of Eq. (40) can be obtained via Fourier transform. Let the steady-state fluctuation $\delta\tilde{\rho}_k(t)$ be:

$$\delta\tilde{\rho}_k(t) = ic \int_{-\infty}^t e^{-Dk^2(t-t')} \eta_k(t') dt'$$

Therefore:

$$\begin{aligned} \langle \delta\tilde{\rho}_k \delta\tilde{\rho}_{-k} \rangle &= -c^2 \int_{-\infty}^t \int_{-\infty}^t e^{-Dk^2(2t-t'-t'')} \langle \eta_k(t') \eta_{-k}(t'') \rangle dt' dt'' \\ &= -c^2 \int_{-\infty}^t e^{-2Dk^2(t-t')} \cdot 2[1 - \cos(k)] dt' \\ &= -c^2 \frac{2[1 - \cos(k)]}{2Dk^2} \\ &= -\frac{c^2[1 - \cos(k)]}{Dk^2} \end{aligned}$$

The final result is:

Structure Factor (Core Result)

$$N_S(k) = \rho_0 - \frac{c^2[1 - \cos(k)]}{Dk^2} = \rho_0 - \frac{c^2}{2D} \left(1 - \frac{k^2}{12} + \dots \right) \quad (45)$$

2.8.6 Physical Analysis

Comparing $S_0 = \lim_{k \rightarrow 0^+} N_S(k) = \rho_0 - c^2/(2D)$ with the Poisson value $S|_{r_1=0} = \rho_0$:

$$\frac{S_0}{\rho_0} = 1 - \frac{c^2}{2D\rho_0} = 1 - \frac{r_1\rho_0}{r_0 + r_1\rho_0} \quad (46)$$

In the interaction-dominated limit $r_1\rho_0 \gg r_0$:

$$\boxed{\frac{S_0}{\rho_0} \rightarrow \frac{1}{2}} \quad (47)$$

This means that long-wavelength density fluctuations are suppressed to one half of the Poisson value!

Key Physical Conclusions

1. When $r_1\rho_0 \gg r_0$ (strong antialigning interaction), $S_0/\rho_0 \rightarrow 1/2$
2. Introducing spontaneous conversion r_0 is equivalent to adding orientational noise, supporting the early assertion that “dynamically established correlations are more pronounced in deterministic models”
3. Lower values (including hyperuniform $S_0 \rightarrow 0$) may be achieved in models with strict center-of-mass or polarization conservation
4. In the $k \rightarrow 0$ limit, the structure factor approaches a **finite constant** rather than zero, so the system is **not strictly hyperuniform**, but exhibits significant fluctuation suppression

2.9 Numerical Verification (One-Dimensional Model)

The paper also reports numerical simulation results for the one-dimensional model. The stochastic process was simulated directly using the Gillespie algorithm¹, with simulation parameters $\lambda = 1$, $r_1 = 1/100$, and variable system size L .

- Simulation results are qualitatively consistent with the predictions of Eq. (45)
- For sufficiently large system sizes, suppression of long-wavelength density fluctuations is observed
- Note that in finite systems, there is a difference between Binomial and Poisson statistics, which requires correction

Remark (Finite-size effects and spurious hyperuniformity). If the offset S_0 is ignored or unknown in the small- k regime, it may lead to erroneous inferences—namely, inferring a spurious power-law behavior. In the case of Fig. 2, this could lead to inferring an “anomalous” scaling of $k^{\approx 0.25}$. This reminds us that inferring hyperuniformity from numerical data without a primary theoretical basis is risky.

3 Two-Dimensional Vicsek-like Model

3.1 Model Definition

To generalize the insights from the one-dimensional analysis to more realistic two-dimensional systems, the paper studies a continuous-time Vicsek-like model¹. This model describes the dynamics of self-driven particles in which the orientational evolution includes antialigning interactions.

3.1.1 Equations of Motion

Denoting the orientation ϑ_i (angle relative to a fixed reference axis) of particle i ($i = 1, \dots, N$), the dynamical equations are:

2D Vicsek-like Model

$$\dot{\mathbf{r}}_i = v_0 \mathbf{n}(\vartheta_i) \quad (48)$$

$$\dot{\vartheta}_i = -\epsilon \sum_{j \in \mathcal{N}_i} \sin(\vartheta_j - \vartheta_i) \quad (49)$$

where:

- $\mathbf{n}(\vartheta) = (\cos \vartheta, \sin \vartheta)^T$ is the unit vector corresponding to direction ϑ
- $\mathcal{N}_i = \{j \mid j \neq i \text{ and } |\mathbf{r}_i - \mathbf{r}_j| \leq R\}$ is the set of neighbors within radius R around particle i
- $\epsilon > 0$ denotes **antialigning** interactions (if negative, it corresponds to conventional aligning interactions)

3.2 Numerical Simulation Methods

3.2.1 Simulation Parameters

The paper integrates the equations of motion using the Heun method (a second-order stochastic differential equation integrator), with parameter settings:

- $\epsilon = v = R = M = 1$ ($M = \rho\pi R^2$ is the expected number of neighbors in the reduced density)
- Particle number N varies from 25 to $2^{12} = 4096$
- Structure factor computed after dimensionless time $t = 500\pi$
- Results averaged over 10^4 samples

3.2.2 Estimate of Critical Noise

The effect of noise can easily be studied numerically. Both positional and orientational noise eventually destroy long-range correlations. Here, **orientational noise** is more relevant, because we are focusing on the effect of aligning interactions.

A rough estimate of the critical noise level can be obtained through the following argument:

1. The typical timescale of interactions is given by the mean free path: $\tau_{\text{free}} \sim R/(Mv)$
2. If active particles move perpendicular to their deterministic direction by about $\Delta \sim R$ due to diffusion during this time, they will interact with different particles
3. This leads to an estimate of the critical orientational diffusion strength: $D_c \approx MvR/\pi$

This is the same scaling as the instability threshold reported in the literature¹.

3.3 Numerical Results

3.3.1 Apparent Hyperuniform Behavior

From direct many-body simulations, the following is observed:

- Seemingly consistent apparent power-law behavior in the reduced structure factor and particle number fluctuations
- However, one must be wary that this may be a **finite-size artifact**
- A description similar to the 1D model prediction (Eq. (45)) can also fit the data

3.3.2 Warning about Spurious Hyperuniformity

The paper particularly emphasizes the following important point:

Risk of inferring hyperuniformity from numerics

Similar apparent hyperuniformity $S \sim k^\alpha$ with “anomalous” (non-integer) exponent $\alpha \approx 0.5$ has been discovered in chiral self-driven particle systems. For the purposes of this paper, the main concern is the **significant reduction** of long-range density fluctuations.

The question of **true hyperuniformity** (i.e., $S(0) = 0$) is best addressed by analytical methods—for the two-dimensional model this remains unresolved at present. The one-dimensional result should serve as a warning in this regard.

3.3.3 Exploration of Parameter Space

For vastly different values of the expected number of interactions $M = \rho\pi R^2$ and dimensionless interaction strength $S_c = \epsilon R/v$, the data show:

- All data show a significant reduction of density fluctuations at large distances

- But the apparent power-law behavior is **non-universal**
- For the parameter set $M = 1, S_c = 1$, the reduced structure factor $\delta S = S - S_0$ can be reasonably explained by k^2 behavior

3.3.4 Collapse of the Effect due to Noise

For sufficiently strong noise, the long-range suppression is destroyed:

- No orientational noise (full circle): significant fluctuation suppression
- Weak noise ($\mu = 0.1$): the effect still persists
- Correlated noise ($\mu = 1$): fluctuation suppression is destroyed, reverting to ordinary behavior

This is consistent with the intuition behind the concept of effective temperature in active matter systems: the self-hybrid dynamics in antialigning models may look similar to the effect of thermal noise at large scales, but the **absence of long-range correlations** is a notable distinction in the case of sufficiently strong explicit noise.

4 Summary and Discussion

4.1 Main Conclusions

This document systematically compiles the theoretical analysis by Boltz and Ihle on density fluctuation suppression in antialigning active matter systems. The core findings include:

1. **One-dimensional analytical solution:** Through the Poisson representation, the structure factor for a one-dimensional active lattice gas model with antialigning interactions is rigorously derived, revealing that long-range density fluctuations are significantly suppressed to approximately 1/2 of the Poisson value
2. **Physical meaning of imaginary noise:** The imaginary noise that naturally emerges in the Poisson representation marks non-Poissonian statistics—this is the mathematical manifestation of the deviation of active systems from equilibrium statistics
3. **Two-dimensional numerical verification:** Apparent hyperuniform behavior is observed in a Vicsek-like model, but it is emphasized that this may be a finite-size effect or a misinterpretation of a finite offset
4. **Warning about numerical inference:** Without a theoretical basis, inferring hyperuniformity from numerical data is unreliable

4.2 Value of the Theoretical Method

Advantages of the Poisson representation in this study:

- Direct, non-perturbative access to fluctuations (the noise term in the final Langevin equation)
- Direct access to the correlations of the noise, which is crucial for determining the structure factor
- In contrast to certain path integral methods (which asymptotically derive Gaussian uncorrelated noise)

4.3 Open Questions

1. **Strict hyperuniformity:** Does true hyperuniformity ($S(0) = 0$) genuinely exist in two-dimensional continuous systems, or is it merely k^2 behavior with a finite offset?
2. **Higher-dimensional generalization:** How does the analytical description of the one-dimensional lattice model generalize to higher dimensions?
3. **Experimental connections:** Comparison with experiments on antialigning self-driven particles (such as the works cited in¹)
4. **Tuning of point pattern statistics:** Antialigning self-driven particles as an independent dynamical process for generating disordered point patterns with tunable statistics

Final Remarks

The transition from Poisson statistics at $v = 0$ (passive system) to non-Poissonian, possibly hyperuniform statistics at $v > 0$ (active system) is a remarkable manifestation of the relevance of long-range correlations even in the absence of direct order.

This suppression of density fluctuations is purely a result of dynamically established correlations (due to aligning interactions), rather than more traditional mechanisms such as elasticity. It is precisely this long-range effect that enables the one-dimensional analytically solvable toy model to capture and connect with the more complex two-dimensional Vicsek-like model.

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